Value of Forecasting Accuracy — Day-Ahead Auction Only

Methodology v1.1: Round-Trip Efficiency, Opportunity-Cost Bidding, and Volume-Balanced Paired Rejection

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1 Introduction

Overview

This document presents the v1.1 mathematical formulation of the CastValue framework: a comparison of the energy arbitrage margins of a Battery Energy Storage System (BESS) in the Day-Ahead Auction (DAA) power market in Germany, under different price forecasts.

v1.1 extends the v1.0 formulation (see `DAA_DE_LU_Formulation.pdf`) with the physical and market frictions a real trader faces:

- **Round-Trip Efficiency (RTE)** of the BESS in the LP energy balance — one MWh delivered to the grid requires more than one MWh purchased from the grid.
- **Opportunity-cost bid derivation** — limit prices for charge bids and discharge asks are derived from the forecast and adjusted for RTE.
- **Out-of-the-money (OTM) rejection** — bids/asks not in the money against the actual clearing price do not fill.
- **Volume-balanced paired rejection** — when OTM fills leave cleared charge and discharge volumes unmatched, additional MTUs are dropped on the oversubscribed side until volumes balance and end-of-day SOC equals start-of-day SOC.
- **Rolling SOC** across days — the day's initial SOC equals the previous day's terminal SOC (the LP is no longer forced to start and end at zero).

The Perfect Foresight benchmark in v1.1 is defined as the LP optimum given RTE, with bids placed at the clearing price (100% fill). It is a "perfect scheduling" benchmark, not a frictionless one — the $(1 - \eta)$ round-trip loss is fully paid in the realised margin.

The v1.1 dataset covers EPEX SPOT DE-LU at native quarter-hourly resolution from 18 October 2025 to 22 February 2026 (126 days). Because the dataset is entirely post the 1 October 2025 market structure change, the hourly coupling constraint of v1.0 is dropped.

This document accompanies the Python implementation of the v1.1 optimization available on our GitHub.

Symbols

Notation extends v1.0. Symbols introduced in v1.1 are flagged in the description column.

Sets and indices

Symbol	Meaning
$D = \{1, \dots, 126\}$	day index (18 Oct 2025 – 22 Feb 2026)
$Q = \{1, \dots, 96\}$	quarter-hour index within the day
$Q^+ = \{1, \dots, 97\}$	quarter-hour index for SOC including terminal state
$\mathcal{K} = \{\mathrm{P}, 1, \dots, k, \mathrm{PF}\}$	price forecast index (Persistence, forecasts $1 \dots k$, Perfect Foresight)
$\hat{\lambda}^{(\mathrm{P})}$	Persistence forecast, where $\hat{\lambda}^{(\mathrm{P})}_{d,q} := \lambda_{d-1,q}$
$\hat{\lambda}^{(\mathrm{PF})}$	Perfect Foresight forecast, where $\hat{\lambda}^{(\mathrm{PF})}_{d,q} := \lambda_{d,q}$

Parameters

Symbol	Meaning	Notes
$\mathtt{cap}^{\mathrm{energy}}$	energy capacity [MWh]	
$\mathtt{cap}^{\mathrm{power}}$	charge/discharge power limit [MW]	
n_{cycles}	allowed daily equivalent full cycles [–]	
$L_d = \mathtt{cap}^{\mathrm{energy}} \cdot n_{\mathrm{cycles}}$	daily energy-throughput limit [MWh]	
$\hat{\lambda}^{\mathrm{DAA,\ k}}_{d,q}$	DAA forecast price from forecast k [EUR/MWh]	
$\lambda^{\mathrm{DAA}}_{d,q}$	DAA actual clearing price [EUR/MWh]	
$\Delta = \frac{1}{4}$	quarter-hour duration [h]	
η	round-trip efficiency [–] ($\eta = 0.85$)	**new in v1.1**

LP decision variables

Symbol	Meaning
$\mathrm{CHA}^{\mathrm{DAA}}_{q,k},\ \mathrm{DIS}^{\mathrm{DAA}}_{q,k} \in [0,1]$	normalized planned charge/discharge in DAA at q, k
$\mathrm{SOC}_{q,k}$	planned state-of-charge at $q \in Q^+$ [MWh]

Derived quantities (post-LP, v1.1)

Symbol	Meaning
$\bar{\lambda}^{\mathrm{cha}}_{d,k},\ \bar{\lambda}^{\mathrm{dis}}_{d,k}$	volume-weighted average forecast charge / discharge price [EUR/MWh]
$B^{\mathrm{cha}}_{q,k},\ B^{\mathrm{dis}}_{q,k}$	charge bid / discharge ask price [EUR/MWh]
$F^{\mathrm{cha}}_{q,k},\ F^{\mathrm{dis}}_{q,k} \in \{0,1\}$	OTM fill indicators
$\widetilde{\mathrm{CHA}}_{q,k},\ \widetilde{\mathrm{DIS}}_{q,k}$	post-OTM accepted volumes [normalized]
$\mathrm{CHA}^{*}_{q,k},\ \mathrm{DIS}^{*}_{q,k}$	realised (delivered) volumes after paired rejection [normalized]
$\mathrm{SOC}^{*}_{q,d,k}$	realised state-of-charge trajectory [MWh]

Outputs

Symbol	Meaning
V_k	Value of forecast k compared with persistence forecast P
C_k	Cost of the error of forecast k compared with perfect foresight PF
$AR_{\text{total},k}$	Total trading margin from DAA across all days $d \in D$, given forecast k
$DR_{d,k}$	Daily trading margin from DAA for day d, given forecast k

2 Calculating Outputs

2.1 Forecast Value

The value (per MW traded BESS) of a price forecast equals the difference in the trading margin generated by trading on that forecast vs. the persistence forecast:

$$V_k = AR_{\text{total},k} - AR_{\text{total},P} \tag{2.1.1}$$

2.2 Cost of Forecast Error

The cost of forecast error equals the gap between Perfect Foresight and forecast k:

$$C_k = AR_{\text{total},PF} - AR_{\text{total},k} \tag{2.2.1}$$

2.3 Total and Daily Trading Margin

$$AR_{\text{total},k} = \sum_{d \in D} DR_{d,k} \tag{2.3.1}$$

The daily trading margin uses *realised* delivered volumes, evaluated at actual prices:

$$DR_{d,k} = \sum_{q \in Q} \lambda^{\text{DAA}}_{d,q} \cdot \texttt{cap}^{\text{power}} \cdot \Delta \cdot \left(\text{DIS}^{*}_{q,d,k} - \text{CHA}^{*}_{q,d,k}\right) \tag{2.3.2}$$

The factor $\texttt{cap}^{\text{power}} \cdot \Delta$ converts normalized values to grid-side MWh per quarter-hour. Note the trader pays for the full grid-side charge volume, including the $(1 - \eta)$ portion lost to RTE.

3 Optimizing Daily Trading Decisions (LP)

3.1 Day-Ahead Auction (DAA) — v1.1

Objective

Maximise expected revenue at forecast prices, given decisions $\text{CHA}^{\text{DAA}}_{q,k}, \text{DIS}^{\text{DAA}}_{q,k}$:

$$\max_{\text{CHA}^{\text{DAA}}_{q,k},\, \text{DIS}^{\text{DAA}}_{q,k}} \sum_{q \in Q} \hat{\lambda}^{\text{DAA}}_{q,k} \cdot \texttt{cap}^{\text{power}} \cdot \Delta \cdot \left(\text{DIS}^{\text{DAA}}_{q,k} - \text{CHA}^{\text{DAA}}_{q,k}\right) \tag{3.1.1}$$

Constraints

SOC bounds:

$$0 \le \text{SOC}_q \le \texttt{cap}^{\text{energy}} \quad \forall q \in Q^{+} \tag{3.1.2}$$

Rolling initial SOC (v1.1 replaces the v1.0 condition $\text{SOC}_1 = 0$):

$$\text{SOC}_{1,d,k} = \text{SOC}^{*}_{97,\,d-1,\,k} \qquad \text{with } \text{SOC}^{*}_{97,\,0,\,k} := 0 \tag{3.1.3}$$

Terminal SOC non-negative but free (v1.1 replaces the v1.0 condition $\mathrm{SOC}_{97} = 0$):

$$\mathrm{SOC}_{97} \geq 0 \tag{3.1.4}$$

RTE-aware energy balance (v1.1 generalises the v1.0 step constraint):

$$\mathrm{SOC}_{q+1} = \mathrm{SOC}_q + \mathtt{cap}^{\mathrm{power}} \cdot \Delta \cdot \left(\eta \cdot \mathrm{CHA}_q^{\mathrm{DAA}} - \mathrm{DIS}_q^{\mathrm{DAA}}\right) \quad \forall q \in Q \tag{3.1.5}$$

i.e. each MWh purchased from the grid stores only η MWh in the battery; each MWh delivered to the grid draws 1 MWh from the battery.

Volume caps (battery-side cycles): the discharge cap remains L_d, and the charge cap is relaxed by $1/\eta$ so the LP can deliver a full discharge cycle despite RTE losses:

$$\mathtt{cap}^{\mathrm{power}} \cdot \Delta \cdot \sum_{q \in Q} \mathrm{CHA}_q^{\mathrm{DAA}} \leq \frac{L_d}{\eta} \tag{3.1.6}$$

$$\mathtt{cap}^{\mathrm{power}} \cdot \Delta \cdot \sum_{q \in Q} \mathrm{DIS}_q^{\mathrm{DAA}} \leq L_d \tag{3.1.7}$$

The hourly coupling constraint of v1.0 is dropped (the v1.1 dataset is entirely 15-minute).

4 Bid Derivation

The trader posts limit prices for each LP-active MTU using a volume-weighted opportunity-cost rule.

4.1 Volume-weighted average forecast prices

Define the LP-planned grid-side energies:

$$E_{q,k}^{\mathrm{cha}} := \mathtt{cap}^{\mathrm{power}} \cdot \Delta \cdot \mathrm{CHA}_{q,k}^{\mathrm{DAA}}, \qquad E_{q,k}^{\mathrm{dis}} := \mathtt{cap}^{\mathrm{power}} \cdot \Delta \cdot \mathrm{DIS}_{q,k}^{\mathrm{DAA}}. \tag{4.1.1}$$

The volume-weighted average forecast charge and discharge prices for day d, forecast k are:

$$\bar{\lambda}_{d,k}^{\mathrm{cha}} = \frac{\sum_{q \in Q} \hat{\lambda}_{q,k}^{\mathrm{DAA}} E_{q,k}^{\mathrm{cha}}}{\sum_{q \in Q} E_{q,k}^{\mathrm{cha}}}, \qquad \bar{\lambda}_{d,k}^{\mathrm{dis}} = \frac{\sum_{q \in Q} \hat{\lambda}_{q,k}^{\mathrm{DAA}} E_{q,k}^{\mathrm{dis}}}{\sum_{q \in Q} E_{q,k}^{\mathrm{dis}}}. \tag{4.1.2}$$

4.2 Bid prices

For LP-active MTUs ($\mathrm{CHA}_{q,k}^{\mathrm{DAA}} > 0$ or $\mathrm{DIS}_{q,k}^{\mathrm{DAA}} > 0$):

$$B_{q,k}^{\mathrm{cha}} = \eta \cdot \bar{\lambda}_{d,k}^{\mathrm{dis}} \qquad \text{if } \mathrm{CHA}_{q,k}^{\mathrm{DAA}} > 0 \tag{4.2.1}$$

$$B_{q,k}^{\mathrm{dis}} = \frac{\bar{\lambda}_{d,k}^{\mathrm{cha}}}{\eta} \qquad \text{if } \mathrm{DIS}_{q,k}^{\mathrm{DAA}} > 0 \tag{4.2.2}$$

Interpretation. The trader pays no more for charging than the expected revenue from later discharging *after* RTE loss; and asks no less for discharging than the cost of the energy charged *gross of* RTE loss. These are the indifference prices implied by the day's forecast schedule.

5 Out-of-the-Money (OTM) Fill Rule

A bid fills only if it is in the money against the actual clearing price:

$$F_{q,k}^{\mathrm{cha}} = \mathbf{1}\left[\, \mathrm{CHA}_{q,k}^{\mathrm{DAA}} > 0 \ \wedge \ B_{q,k}^{\mathrm{cha}} \geq \lambda_{q,d}^{\mathrm{DAA}} \,\right] \tag{5.0.1}$$

$$F_{q,k}^{\mathrm{dis}} = \mathbf{1}\left[\, \mathrm{DIS}_{q,k}^{\mathrm{DAA}} > 0 \ \wedge \ B_{q,k}^{\mathrm{dis}} \leq \lambda_{q,d}^{\mathrm{DAA}} \,\right] \tag{5.0.2}$$

Post-OTM accepted volumes:

$$\widetilde{\mathrm{CHA}}_{q,k} = F_{q,k}^{\mathrm{cha}} \cdot \mathrm{CHA}_{q,k}^{\mathrm{DAA}}, \qquad \widetilde{\mathrm{DIS}}_{q,k} = F_{q,k}^{\mathrm{dis}} \cdot \mathrm{DIS}_{q,k}^{\mathrm{DAA}}. \tag{5.0.3}$$

6 Volume-Balanced Paired Rejection

When forecast errors lead to asymmetric OTM fills, cleared charge and discharge volumes diverge. To enforce energy balance and avoid open positions at end-of-day, the worst-margin MTUs on the oversubscribed side are progressively rejected until volumes match.

Define the cleared battery-side volumes after OTM:

$$V_{d,k}^{\text{cha-stored}} = \texttt{cap}^{\text{power}} \cdot \Delta \cdot \eta \cdot \sum_{q \in Q} \widetilde{\text{CHA}}_{q,k} \tag{6.0.1}$$

$$V_{d,k}^{\text{dis-drawn}} = \texttt{cap}^{\text{power}} \cdot \Delta \cdot \sum_{q \in Q} \widetilde{\text{DIS}}_{q,k} \tag{6.0.2}$$

Procedure. Initialise $\text{CHA}^*_{q,k} \leftarrow \widetilde{\text{CHA}}_{q,k}$ and $\text{DIS}^*_{q,k} \leftarrow \widetilde{\text{DIS}}_{q,k}$.

1. If $V_{d,k}^{\text{cha-stored}} > V_{d,k}^{\text{dis-drawn}}$: identify cleared charge MTUs and sort by descending actual price $\lambda_{q,d}^{\text{DAA}}$ (most-expensive purchase first = worst margin first). Sequentially reduce $\text{CHA}^*_{q,k}$ until battery-side volumes match.
2. If $V_{d,k}^{\text{dis-drawn}} > V_{d,k}^{\text{cha-stored}}$: identify cleared discharge MTUs and sort by ascending actual price (cheapest sale first = worst margin first). Sequentially reduce $\text{DIS}^*_{q,k}$ until battery-side volumes match.
3. Otherwise ($V_{d,k}^{\text{cha-stored}} = V_{d,k}^{\text{dis-drawn}}$): no adjustment needed.

The realised SOC trajectory is then computed analogously to (3.1.5), using the realised volumes:

$$\text{SOC}^*_{q+1,d,k} = \text{SOC}^*_{q,d,k} + \texttt{cap}^{\text{power}} \cdot \Delta \cdot \left(\eta \cdot \text{CHA}^*_{q,k} - \text{DIS}^*_{q,k}\right) \quad \forall q \in Q \tag{6.0.3}$$

with $\text{SOC}^*_{1,d,k} = \text{SOC}^*_{97,d-1,k}$. By construction $\text{SOC}^*_{97,d,k} = \text{SOC}^*_{1,d,k}$ — daily energy balances, and there is no carryover or imbalance position.

7 Perfect Foresight Benchmark

Under Perfect Foresight ($\hat{\lambda}^{\text{DAA}}_{q,PF} = \lambda^{\text{DAA}}_{q,d}$), the trader knows the clearing price exactly and posts limit prices at clearing:

$$B^{\text{cha}}_{q,PF} = B^{\text{dis}}_{q,PF} = \lambda^{\text{DAA}}_{q,d}. \tag{7.0.1}$$

All LP-scheduled MTUs are then in the money ($F^{\text{cha}}_{q,PF} = F^{\text{dis}}_{q,PF} = 1$ wherever the LP is active), so:

$$\text{CHA}^*_{q,PF} = \text{CHA}^{\text{DAA}}_{q,PF}, \qquad \text{DIS}^*_{q,PF} = \text{DIS}^{\text{DAA}}_{q,PF}. \tag{7.0.2}$$

Paired rejection is a no-op. The Perfect Foresight margin equals the LP optimum subject to RTE — not a frictionless ceiling. The $(1-\eta)$ round-trip loss is fully reflected in (2.3.2).

8 Summary of Differences from v1.0

Aspect	v1.0	v1.1
RTE in LP	none (1:1 efficiency)	$\eta = 0.85$ on charge side
Charge-side volume cap	L_d	L_d/η
Initial SOC	$\text{SOC}_1 = 0$	rolling: $\text{SOC}_{1,d} = \text{SOC}^*_{97,d-1}$
Terminal SOC	$\text{SOC}_{97} = 0$	$\text{SOC}_{97} \geq 0$ (free)
Hourly coupling	active before 1 Oct 2025	dropped (15-min only)
Bid pricing	implicit (zero-cap, infinite-floor)	opportunity-cost, RTE-adjusted
Bid acceptance	100% (guaranteed)	OTM check vs. actual
Volume reconciliation	none (LP guarantees by $\text{SOC}_{97} = 0$)	paired rejection
Perfect Foresight	frictionless LP optimum	RTE-aware LP optimum, bids at clearing